CP2.1 — Complex Numbers: Further Topics

1. Exponential (Euler) Form

Euler's Formula: For any real θ,

$$e^{i\theta} = \cos\theta + i\sin\theta$$

A complex number $z = r(\cos\theta + i\sin\theta)$ can therefore be written in *exponential form* as $z = re^{i\theta}$.

Key special cases worth memorising:

- $e^{i\pi} + 1 = 0$ — Euler's identity
- $e^{i\pi/2} = i$, $e^{i\pi} = -1$, $e^{3i\pi/2} = -i$
- $e^{-i\theta} = \cos\theta - i\sin\theta$

From Euler's formula we derive two extremely useful identities:

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

Exam tip: Exponential form makes multiplication and division trivial. $z_1 z_2 = r_1 r_2 e^{i(\theta_1+\theta_2)}$ — just multiply moduli and add arguments.

2. De Moivre's Theorem

De Moivre's Theorem: For any integer n,

$$\big(r(\cos\theta + i\sin\theta)\big)^n = r^n(\cos n\theta + i\sin n\theta)$$

Equivalently, $\left(re^{i\theta}\right)^n = r^n e^{in\theta}$.

2.1 Proof by Mathematical Induction (positive integers)

Base case $n = 1$: $r^1(\cos 1 \cdot \theta + i \sin 1 \cdot \theta) = r(\cos\theta + i\sin\theta)$. True.

Inductive step: Assume true for $n = k$, i.e. $(r(\cos\theta + i\sin\theta))^k = r^k(\cos k\theta + i\sin k\theta)$. Then:

$$\left(r(\cos\theta + i\sin\theta)\right)^{k+1} = r^k(\cos k\theta + i\sin k\theta) \cdot r(\cos\theta + i\sin\theta)$$

$$= r^{k+1}\left[(\cos k\theta\cos\theta - \sin k\theta\sin\theta) + i(\cos k\theta\sin\theta + \sin k\theta\cos\theta)\right]$$

$$= r^{k+1}\left(\cos(k+1)\theta + i\sin(k+1)\theta\right)$$

using the compound angle formulas. So the result holds for $n = k + 1$. By induction, true for all positive integers.

Extension to negative integers: For $n = -m$ where $m > 0$:

$$z^{-m} = \frac{1}{r^m(\cos m\theta + i\sin m\theta)} = \frac{\cos m\theta - i\sin m\theta}{r^m} = r^{-m}(\cos(-m\theta) + i\sin(-m\theta))$$

Exam tip: If asked to prove de Moivre's theorem, you must use induction for positive integers, then extend to negative integers by taking reciprocals. Don't skip the base case or the algebraic step using compound angles.

3. Applications — cos nθ and sin nθ in Terms of Powers

Using de Moivre, $\cos n\theta + i\sin n\theta = (\cos\theta + i\sin\theta)^n$. Expand the RHS with the binomial theorem, then equate real and imaginary parts.

Example: Express $\cos 3\theta$ and $\sin 3\theta$ in terms of $\cos\theta$ and $\sin\theta$.

Let $c = \cos\theta$, $s = \sin\theta$. Then:

$$(c+is)^3 = c^3 + 3c^2(is) + 3c(is)^2 + (is)^3 = c^3 + 3ic^2s - 3cs^2 - is^3$$
$$= (c^3 - 3cs^2) + i(3c^2s - s^3)$$

Therefore:

$$\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta = 4\cos^3\theta - 3\cos\theta$$
$$\sin 3\theta = 3\cos^2\theta\sin\theta - \sin^3\theta = 3\sin\theta - 4\sin^3\theta$$

4. Expressing $\cos^n\theta$ and $\sin^n\theta$ in Terms of Multiple Angles

Let $z = e^{i\theta} = \cos\theta + i\sin\theta$, so $z^{-1} = e^{-i\theta} = \cos\theta - i\sin\theta$.

$$z + z^{-1} = 2\cos\theta, \qquad z - z^{-1} = 2i\sin\theta$$
$$z^n + z^{-n} = 2\cos n\theta, \qquad z^n - z^{-n} = 2i\sin n\theta$$

Example: Express $\cos^4\theta$ in terms of multiple angles.

$(2\cos\theta)^4 = (z+z^{-1})^4$. Expand by binomial:

$$(z+z^{-1})^4 = z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4} = (z^4+z^{-4}) + 4(z^2+z^{-2}) + 6$$
$$= 2\cos 4\theta + 8\cos 2\theta + 6$$

So $16\cos^4\theta = 2\cos 4\theta + 8\cos 2\theta + 6$, giving:

$$\cos^4\theta = \frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8}$$

Example: Express $\sin^5\theta$ in terms of multiple angles.

$(2i\sin\theta)^5 = (z-z^{-1})^5$. Note $(2i)^5 = 32i^5 = 32i$.

$$(z-z^{-1})^5 = z^5 - 5z^3 + 10z - 10z^{-1} + 5z^{-3} - z^{-5}$$
$$= (z^5 - z^{-5}) - 5(z^3 - z^{-3}) + 10(z - z^{-1})$$
$$= 2i\sin 5\theta - 10i\sin 3\theta + 20i\sin\theta$$

So $32i\sin^5\theta = 2i(sin5\theta - 5\sin 3\theta + 10\sin\theta)$, giving:

$$\sin^5\theta = \frac{1}{16}(\sin 5\theta - 5\sin 3\theta + 10\sin\theta)$$

Exam tip: These reductions are very useful for integrating powers of sin/cos. For $\cos^n\theta$, use $(z+z^{-1})^n$. For $\sin^n\theta$, use $(z-z^{-1})^n$. Remember the factor of $(2i)^n$ for sin.

5. nth Roots of a Complex Number

Finding nth roots: To solve $z^n = w$ where $w = Re^{i\Phi}$:
Write $z = re^{i\theta}$. Then $r^n e^{in\theta} = Re^{i\Phi}$.
So $r = R^{1/n}$ and $n\theta = \Phi + 2k\pi$ for $k = 0, 1, 2, \ldots, n-1$.

The n distinct roots are:

$$z_k = R^{1/n} \exp\left(i\frac{\Phi + 2k\pi}{n}\right), \quad k = 0, 1, \ldots, n-1$$

Key geometric fact: The n roots lie on a circle of radius $R^{1/n}$ centred at the origin, equally spaced at angular intervals of $\dfrac{2\pi}{n}$.

Example: Find all cube roots of $8i$.

Write $8i = 8e^{i\pi/2}$. So $R = 8$, $\Phi = \pi/2$, $n = 3$.
$r = 8^{1/3} = 2$. Arguments: $\theta_k = \dfrac{\pi/2 + 2k\pi}{3}$.

- $k = 0$: $\theta = \pi/6$, so $z_0 = 2e^{i\pi/6} = 2(\frac{\sqrt{3}}{2} + \frac{i}{2}) = \sqrt{3} + i$
- $k = 1$: $\theta = \pi/6 + 2\pi/3 = 5\pi/6$, so $z_1 = 2e^{5i\pi/6} = -\sqrt{3} + i$
- $k = 2$: $\theta = \pi/6 + 4\pi/3 = 3\pi/2$, so $z_2 = 2e^{3i\pi/2} = -2i$

6. nth Roots of Unity

The **nth roots of unity** are the solutions of $z^n = 1 = e^0$. They are:

$$\omega_k = e^{2\pi ik/n} = \cos\frac{2k\pi}{n} + i\sin\frac{2k\pi}{n}, \quad k = 0, 1, \ldots, n-1$$

Writing $\omega = e^{2\pi i/n}$, the roots are $1, \omega, \omega^2, \ldots, \omega^{n-1}$.

6.1 Key Properties of nth Roots of Unity

- **Sum is zero:** $1 + \omega + \omega^2 + \cdots + \omega^{n-1} = 0$
- **Product:** $\omega^n = 1$ for all roots
- **Conjugate pairs:** $\overline{\omega^k} = \omega^{n-k}$
- The roots form a regular n-gon on the unit circle

Proof that sum = 0: The roots are a geometric series with first term 1 and ratio $\omega \neq 1$:

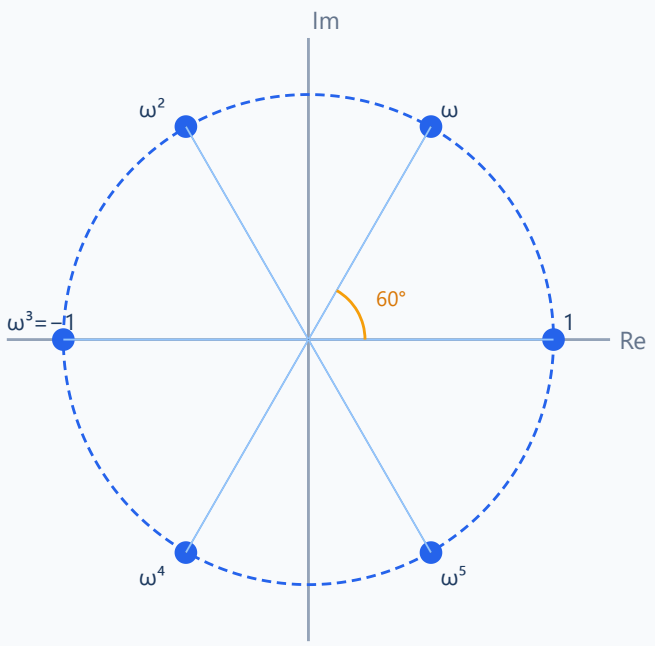
$$\sum_{k=0}^{n-1} \omega^k = \frac{1-\omega^n}{1-\omega} = \frac{1-1}{1-\omega} = 0$$

Example: Find the 5th roots of unity and show their sum is zero.

$\omega = e^{2\pi i/5}$. Roots: $1, \omega, \omega^2, \omega^3, \omega^4$.
These lie on the unit circle at angles $0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ$.
Sum: $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$ (geometric series or from sum of roots of $z^5 - 1 = 0$, coefficient of z^4 is 0).

Exam tip: The sum of all nth roots of unity equals zero. This is because they are roots of $z^n - 1 = 0$, and by Vieta's formulas, the sum of roots equals the negative of the coefficient of z^{n-1}, which is 0.

6.2 Diagram — 6th Roots of Unity



6th Roots of Unity: ω = e^(2πi/6), spaced 60° apart

7. Solving $z^n = w$ — Method Summary

Step	Action
1	Write w in modulus-argument form: $w = Re^{i\Phi}$ where $-\pi < \Phi \leq \pi$
2	The modulus of each root: $r = R^{1/n}$
3	Arguments: $\theta_k = \dfrac{\Phi + 2k\pi}{n}$ for $k = 0, 1, \ldots, n-1$
4	Write each root: $z_k = r\left(\cos\theta_k + i\sin\theta_k\right)$
5	Simplify, giving exact values where possible

Exam tip: Always choose $k = 0, 1, \ldots, n-1$ (not negative values) to get n distinct roots. Keep arguments in $(-\pi, \pi]$ if asked for principal arguments by adjusting by 2π.

8. Geometric Applications of Roots

When the n roots of $z^n = w$ are plotted on an Argand diagram they form a regular n-gon inscribed in a circle of radius $|w|^{1/n}$ centred at the origin.

Example: The 4th roots of -16.

$-16 = 16e^{i\pi}$. Radius $= 16^{1/4} = 2$. Arguments $= \dfrac{\pi + 2k\pi}{4}$ for $k = 0, 1, 2, 3$.
$k = 0$: $\theta = \pi/4 \Rightarrow z_0 = 2e^{i\pi/4} = \sqrt{2} + i\sqrt{2}$
$k = 1$: $\theta = 3\pi/4 \Rightarrow z_1 = -\sqrt{2} + i\sqrt{2}$
$k = 2$: $\theta = 5\pi/4 \Rightarrow z_2 = -\sqrt{2} - i\sqrt{2}$
$k = 3$: $\theta = 7\pi/4 \Rightarrow z_3 = \sqrt{2} - i\sqrt{2}$
These form a square of side $2\sqrt{2}$, centred at the origin.

9. Using de Moivre for Trigonometric Identities

9.1 Strategy for cos nθ or sin nθ in terms of cos/sin

1. Write $\cos n\theta + i\sin n\theta = (\cos\theta + i\sin\theta)^n$
2. Expand RHS by binomial theorem
3. Equate real parts for $\cos n\theta$, imaginary parts for $\sin n\theta$
4. Replace $\sin^2\theta$ with $1 - \cos^2\theta$ if needed (or vice versa)

9.2 Strategy for $\cos^n\theta$ or $\sin^n\theta$ in terms of multiple angles

1. Use $2\cos\theta = z + z^{-1}$ or $2i\sin\theta = z - z^{-1}$
2. Raise to power n and expand by binomial
3. Group terms as $z^k + z^{-k} = 2\cos k\theta$ or $z^k - z^{-k} = 2i\sin k\theta$

Example (Integration application): Find $\int \sin^4\theta\, d\theta$.

First express $\sin^4\theta$ in multiple angles:

$$(2i\sin\theta)^4 = (z - z^{-1})^4 \Rightarrow 16\sin^4\theta = z^4 - 4z^2 + 6 - 4z^{-2} + z^{-4}$$
$$= 2\cos 4\theta - 8\cos 2\theta + 6$$

So $\sin^4\theta = \frac{1}{8}(\cos 4\theta - 4\cos 2\theta + 3)$.

$$\int \sin^4\theta\, d\theta = \frac{1}{8}\left(\frac{\sin 4\theta}{4} - 2\sin 2\theta + 3\theta\right) + C$$

Exam tip: This technique is the key to integrating even powers of sin and cos — it reduces them to integrals of multiple angles which are trivial.

10. Summary Table

Concept	Formula
Euler form	$e^{i\theta} = \cos\theta + i\sin\theta$

De Moivre (integer n)	$(re^{i\theta})^n = r^n e^{in\theta}$
cos from z	$2\cos\theta = z + z^{-1}$, $2\cos n\theta = z^n + z^{-n}$
sin from z	$2i\sin\theta = z - z^{-1}$, $2i\sin n\theta = z^n - z^{-n}$
nth roots of w	$z_k = \lvert w\rvert^{1/n} e^{i(\arg w + 2k\pi)/n}$, $k = 0, \ldots, n-1$
nth roots of unity	ω^k where $\omega = e^{2\pi i/n}$; sum = 0
Sum of roots of unity	$1 + \omega + \omega^2 + \cdots + \omega^{n-1} = 0$

11. Exam-Style Practice Questions

1. **[Proof]** Use mathematical induction to prove de Moivre's theorem for positive integers: $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.
2. **[Roots]** Solve $z^4 = -4$, giving your answers in the form $a + bi$ where $a, b \in \mathbb{R}$. Show the roots on an Argand diagram.
3. **[Trig identity]** (a) Using de Moivre's theorem, show that $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$. (b) Hence solve $16c^5 - 20c^3 + 5c = \frac{1}{\sqrt{2}}$ for c where $0 < c < 1$.
4. **[Multiple angles]** Express $\sin^6\theta$ in terms of cosines of multiples of θ. Hence evaluate $\int_0^{\pi/2} \sin^6\theta \, d\theta$.
5. **[Roots of unity]** (a) Write down the 5th roots of unity $1, \omega, \omega^2, \omega^3, \omega^4$. (b) Show that $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$. (c) Find the value of $(1+\omega)(1+\omega^2)(1+\omega^3)(1+\omega^4)$.

General exam tips for CP2.1:

- Always quote/state de Moivre's theorem before using it
- For root-finding, write the equation in form $z^n = Re^{i\Phi}$ first — don't try to work in $a + bi$ form
- Diagrams: mark modulus of each root and angle spacing clearly
- Check: if roots of unity sum to 0, your values are likely correct
- Know Euler's identity $e^{i\pi} = -1$ — it occasionally appears in "show that" questions